

PROJECT PROPOSAL

Title

Web-Based Citizen Complaint and Service Request Management System for Adama City Administration

1. Background of the Study

Adama City Administration provides various municipal services to citizens. Citizens often need to submit complaints regarding public services such as road maintenance, waste management, water supply, street lighting, and other community-related issues. In addition, citizens may request different municipal services from the city administration.

Currently, many complaints and service requests are handled manually through paper-based processes or office visits. This can lead to delays, poor record management, limited tracking capabilities, and reduced transparency.

To address these challenges, a web-based system is proposed to allow citizens to submit complaints and service requests online, track their progress, and receive feedback from responsible departments.

2. Problem Statement

The existing complaint and service request management process faces several challenges:

- Manual handling of complaints and requests
- Difficulty in tracking complaint status
- Delayed response from responsible departments
- Lack of centralized data storage
- Poor communication between citizens and city administration
- Limited reporting and monitoring capabilities
- Inefficient record management

3. General Objective

To develop a web-based citizen complaint and service request management system for Adama City Administration.

4. Specific Objectives

- To provide online citizen registration and authentication
- To enable citizens to submit complaints electronically
- To enable citizens to submit service requests electronically
- To provide complaint and request tracking functionality
- To provide an administrative dashboard
- To improve communication between citizens and city administration
- To generate reports and statistics
- To maintain centralized records of complaints and requests

5. Scope of the Project

The system will focus on:

- User registration and login
- Complaint submission
- Service request submission
- Complaint tracking
- Service request tracking
- User management
- Status management
- Report generation
- Dashboard management

The system will not include online payment services or advanced GIS mapping functionalities.

6. Proposed System

The proposed system will be a web-based application that allows citizens to access services through the internet.

Citizens can:

- Create accounts
- Login to the system
- Submit complaints
- Submit service requests
- View complaint status
- Receive updates

Administrators can:

- Manage users
- View complaints and requests
- Assign complaints to departments
- Update statuses
- Generate reports

Department officers can:

- View assigned complaints
- Process requests
- Update progress
- Mark requests as resolved

7. Actors of the System

1. Citizen

Responsibilities:

- Register account
- Login
- Submit complaint
- Submit service request
- View status
- Update profile

2. Administrator

Responsibilities:

- Manage users
- Manage complaints
- Manage service requests
- Assign requests
- Generate reports
- Monitor system activities

3. Department Officer

Responsibilities:

- View assigned complaints
- Process complaints
- Process service requests
- Update progress
- Mark tasks as completed

8. Functional Requirements

The system shall provide:

- User registration
- User login
- Password management
- Complaint submission
- Service request submission
- Status tracking
- Dashboard management
- User management
- Complaint assignment
- Report generation
- Search functionality
- Notification functionality

9. Non-Functional Requirements

Security

- Password encryption
- User authentication
- Role-based access control

Performance

- Fast response time
- Efficient data retrieval

Reliability

- Consistent operation
- Data backup support

Usability

- User-friendly interface
- Easy navigation

Scalability

- Ability to support increasing users

10. System Workflow

Step 1: Citizen registers an account.

Step 2: Citizen logs into the system.

Step 3: Citizen submits a complaint or service request.

Step 4: The system stores the information in the database.

Step 5: Administrator reviews the submission.

Step 6: Administrator assigns the request to the appropriate department.

Step 7: Department officer processes the request.

Step 8: Status is updated:

- Pending
- In Progress
- Resolved

Step 9: Citizen views progress through the system.

11. Technology Stack

Frontend

- React.js
- HTML5
- CSS3
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB

Authentication

- JWT (JSON Web Token)

Development Tools

- Visual Studio Code
- Postman
- Git
- GitHub

12. Database Design

Users Collection

Fields:

- userId
- fullName
- email
- password
- role
- phoneNumber

Complaints Collection

Fields:

- complaintId
- title
- description
- category
- status
- citizenId

- assignedDepartment
- createdDate

Service Requests Collection

Fields:

- requestId
- serviceType
- description
- status
- citizenId
- assignedDepartment
- createdDate

13. Expected Benefits

For Citizens

- Easy complaint submission
- Faster service access
- Transparency in service delivery
- Real-time status tracking

For Adama City Administration

- Improved record management
- Better monitoring and control
- Faster response to citizens
- Improved service quality
- Better reporting and decision-making

14. Expected Outcome

The project will produce a fully functional web-based system that enables efficient management of citizen complaints and service requests. The system will improve communication between citizens and Adama City Administration while increasing transparency, efficiency, and accountability.

15. Tools for Implementation

- Visual Studio Code
- Node.js
- Express.js
- MongoDB
- React.js
- Postman
- GitHub

16. Future Enhancements

- SMS notifications

- Email notifications
- Mobile application
- GIS integration
- Analytics dashboard
- Multi-language support